

四川大学期末考试试题（闭卷） - 评分标准

（2023——2024 学年第 1 学期） A 卷

课程号：304032030 课序号：1-8 课程名称：计算机网络（双语） 成绩：
任课教师：陈黎、边昂、傅静涛、杨朝斌、朱敏、张靖宇 印题份数：
适用专业年级：2021 学生人数： 学号： 姓名：

考生承诺

我已认真阅读并知晓《四川大学考场规则》和《四川大学本科学生考试违纪作弊处分规定（修订）》，郑重承诺：

- 1、已按要求将考试禁止携带的文具用品或与考试有关的物品放置在指定地点；
- 2、不带手机进入考场；
- 3、考试期间遵守以上两项规定，若有违规行为，同意按照有关条款接受处理。

考生签名：

I. Acronyms (10 points, 1 point for each acronym)

Consider the following acronyms: AIMD, ARP, BGP, CDMA, CIDR, CNAME, DHCP, DNS, FDM, FIN, FTP, HTTP, ICANN, MSS, MTU, LAN, NAT, RFC, RIP, RST, SYN, TDM, TLD, UDP, URL, WAN.

Match the acronyms to the descriptions below, using each acronym exactly once. Supply exactly one answer for each item below. In some cases, an acronym might fit more than one question, but there is only one way to answer these so that every acronym is used exactly once. **BGP, MSS, URL, CNAME, ARP, FTP, RFC, LAN, HTTP, TDM**

1. A protocol used in IP routing to exchange routing information between different Autonomous Systems (AS) in the Internet.
2. The largest amount of data that can be encapsulated within a TCP segment
3. A standardized address format used to specify the location of a resource on the Internet.
4. A type of DNS record that maps an alias or subdomain to the canonical (true) domain name.
5. A network protocol used to map an IP address to a corresponding MAC address.
6. A protocol used for transferring files between a client and a server on a network.

7. Documents used to define standards, protocols, and guidelines for the Internet community.
8. A network that connects computers and devices within a limited geographical area, such as a home or campus.
9. A protocol used for transmitting hypertext documents on the World Wide Web.
10. A technique that allows multiple signals to be transmitted over a single communication channel by dividing the channel into time slots

II. Single Choice (30 points, each 2 points)

A,C,C,B,B, C,B,B,D,A, A,C,C,A,A

1. Which DNS record type is used to associate an IP address with a domain name?
A) A record
B) MX record
C) CNAME record
D) NS record
2. What does an HTTP 304 response code indicate?
A) The requested resource has been permanently moved to a new location.
B) The server encountered an error while processing the request.
C) The requested resource has not been modified since the last request.
D) The server is temporarily unavailable or overloaded.
3. Which type of applications or services are commonly associated with UDP usage?
A) File transfer and email services
B) Web browsing and HTTP traffic
C) Video streaming and online gaming
D) Remote shell access and Telnet
4. What happens when the receiver's buffer is full during TCP flow control?
A) The receiver discards incoming data packets.
B) The sender reduces its data transmission rate.
C) The receiver sends a congestion notification to the sender.

- D) The sender increases the size of the sliding window.
5. A routing change has the potential to adversely affect which of the following TCP mechanisms.
- A) Flow control
 - B) Congestion control
 - C) Initial Sequence Number exchange
 - D) Demultiplexing
6. What information is typically included in a DHCP offer?
- A) only IP address.
 - B) MAC address, port number, and lease duration.
 - C) IP address, subnet mask, default gateway, and DNS server addresses.
 - D) Network topology and routing table information.
7. Which of the following statements about default gateways is **incorrect**?
- A) A default gateway is used by devices to send traffic to destinations outside their local network.
 - B) The default gateway typically has the highest IP address in the local network.
 - C) A default gateway should be specified on each device on a network for proper routing.
 - D) The default gateway is usually the IP address of the router interface connecting to the local network.
8. The 10.0.0.0/8 network is twice as big as 10.0.0.0/16 in terms of the possible number of hosts in the network.
- A) True
 - B) False
9. Which IP address range is reserved for private networks and cannot be routed on the public internet?
- A) 10.0.0.0 - 10.255.255.255
 - B) 172.16.0.0 - 172.31.255.255
 - C) 192.168.0.0 - 192.168.255.255

D) All of the above

10. Which of the following is a characteristic of the link layer?

- A) End-to-end flow control
- B) Network address translation (NAT)
- C) Routing table maintenance
- D) Media access control (MAC)

11. Which of the following statements describes a key difference between the Go-Back-N (GBN) and Selective Repeat (SR) protocols?

- A) GBN retransmits all unacknowledged packets, while SR retransmits only the lost packets.
- B) GBN retransmits only the lost packets, while SR retransmits all unacknowledged packets.
- C) GBN uses a sliding window for flow control, while SR uses a stop-and-wait mechanism.
- D) GBN supports out-of-order delivery, while SR ensures in-order delivery.

12. Which network scenario is most likely to experience slower convergence in the Distance Vector algorithm?

- A) Large-scale network with low traffic load
- B) Small-scale network with high traffic load
- C) Network with frequent link failures and topology changes
- D) Network with well-balanced link bandwidth distribution

13. What is the purpose of self-learning in a switch?

- A) To automatically configure VLANs on the switch
- B) To dynamically assign IP addresses to connected devices
- C) To learn and update the MAC addresses of devices connected to the switch
- D) To optimize the routing path between switches

14. Given a 7-bit generator polynomial, $G=10011$, assuming that the value of D (Data) is 1010100000, what is the value of R (CRC bits) ?

- A) 1001

- B) 1111
- C) 0000
- D) 0100

15. The characteristics of the UDP protocol include:

- A) Detection of bit errors
- B) Congestion control
- C) Flow control
- D) Bandwidth guarantee

III. Comprehensive problems (60 points)

1. Suppose there is a 10 Mbps microwave link between a geostationary satellite and its base station on Earth. **Every minute** the satellite takes a digital photo and sends it to the base station. Assume a propagation speed of $2.4 \cdot 10^8$ meters/sec. The geostationary satellite is 36,000 kilometers away from earth surface.

- A) What is the propagation delay of the link? (5 points)
- B) What is the bandwidth-delay product, $R \cdot \text{Delay}_{prop}$? (5 points)
- C) Let x denote the size of the photo. What is the minimum value of x for the microwave link to be continuously transmitting? (5 points)

- a) 150 msec
- b) 1,500,000 bits
- c) 600,000,000 bits

2. Please answer the following questions about the TCP/IP protocol stack.

- A) What are the five layers in the Internet protocol stack? (3 points)
- B) What are the principal responsibilities of each of these layers? (3 points)
- C) What is the purpose of the Network Address Translation (NAT) function at the network layer? (2 points)

- A) App, transport, network, link, physical
- B) Businessss layer, delivery pkt between processes, delivery pkt between host, transfer packet among neighbor nodes, signal handling
- C) 数据包通过路由器或防火墙时, private IP 地址被映射至单个 public IP, 安全性和减少对公网 IP 需求。

3. Recall that with the CSMA/CD protocol, the adapter waits $K \cdot 512$ bit times after a collision, where K is drawn randomly. For $K = 100$,

A) What will be the backoff time for the node to retry carrier sensing in a 10Mbps network? (6 points)

B) What will be the backoff time for the node to retry carrier sensing in a 100Mbps network? (6 points)

A) $51.2 \cdot 10^3 \text{ bits} / 10 \cdot 10^6 \text{ bps} = 5.12 \text{ msec}$ 或 0.00512 s 或 $5120 \mu\text{s}$

B) $51.2 \cdot 10^3 \text{ bits} / 100 \cdot 10^6 \text{ bps} = 0.512 \text{ msec}$ 或 0.000512 s 或 $512 \mu\text{s}$

4. Consider a data transfer scenario where a browser process, referred to as client, initiates communication with a web server process, referred to as server. Client is configured with an initial TCP receive window size of 1024 bytes, while server is configured with an initial TCP receive window size of 512 bytes. Assume that TCP initial sequence numbers start at 0 for this problem.

A) In this scenario, the client has received data up to byte 560 from the server, but its browser has only processed the first 60 bytes. What will be the ACK number and receive window size that the client will include in the next TCP header sent to the server. (5 points)

B) Later in the same connection, the server sets its congestion window to 1 MSS (536 bytes) and receives an advertised flow-control window of 560 bytes from the client. The last acknowledged byte received by the server from the client is byte 700, and the last byte sent by the server to client is byte 900.

a) What is the smallest byte number that client will not accept? (5 points)

b) Assuming that server doesn't receive any more ACKs and its window does not change, what is the greatest byte number that server can send? (5 points)

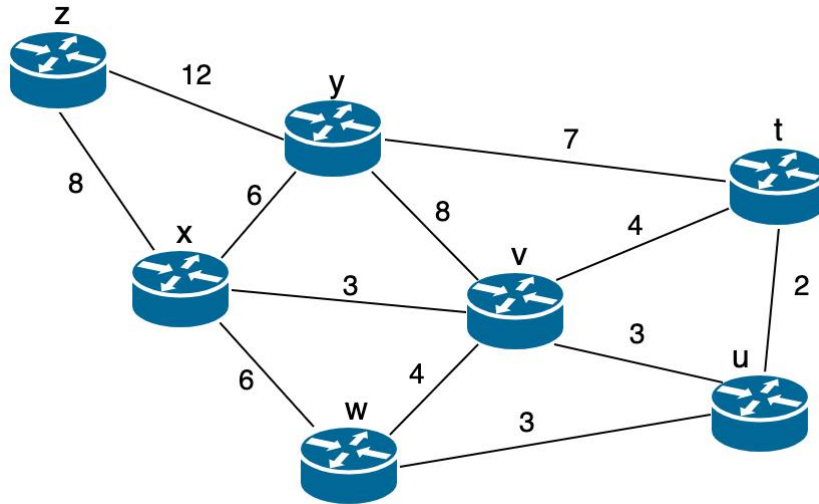
A) SEQ Number = 561 ; Window = $1024 - 500 = 524$

a) 还能继续接收 $560 - 200 = 360$ 个字节。 因此最多能接收到 $900 + 360 = 1260 \text{ byte}$, 因此客户端第一个不可能接收的字节是 byte 1261

b) 发送窗口由 $\min(\text{cwnd}, \text{rwnd})$ 决定。当前 $\text{svr cwnd} = 536$, 其中有 200 字节已送出还未被 ack , 因此还可以发送 $536 - 200 = 336$ 字节

svr 最大可以发送的字节是: $\min(1236, 1260) = 1236$

5. Consider the given network topology. Apply Dijkstra's shortest-path algorithm to determine the shortest path from the source node y to all other nodes in the network. Present the step-by-step execution of the algorithm by creating a table similar to the one discussed during the class. (10 points)



Step	N'	$D(x), p(x)$	$D(u), p(u)$	$D(v), p(v)$	$D(w), p(w)$	$D(t), p(t)$	$D(z), p(z)$
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Step	N'	$D(x), p(x)$	$D(u), p(u)$	$D(v), p(v)$	$D(w), p(w)$	$D(t), p(t)$	$D(z), p(z)$
	y	6,y	∞	8,y	∞	7,y	12,y
	yx	6,y	∞	8,y	12,x	7,y	12,y
	yxt	6,y	9,t	8,y	12,x	7,y	12,y
	yxtv	6,y	9,t	8,y	12,x	7,y	12,y
	yxtvu	6,y	9,t	8,y	12,x	7,y	12,y
	yxtvuw	6,y	9,t	8,y	12,x	7,y	12,y
	yxtvu wz	6,y	9,t	8,y	12,x	7,y	12,y